Uniformly Automatic Classes of Finite Groups of Bounded Rank-Width

Faried Abu Zaid ✉
Independent Researcher, Germany

Abstract

Finite nilpotent groups of class 2 and exponent p correspond, by the Baer correspondence, to alternating bilinear maps over $\mathbb{F}_p$; they dominate the enumeration of finite groups, and their isomorphism problem is equivalent to isometry of alternating matrix spaces and complete for the tensor-isomorphism class. We identify a structural parameter of such groups under which they become uniformly tame for automata-based methods: the *cut-rank* of the commutation form over a linear or tree layout, i.e. its $\mathbb{F}_p$ rank-width. For every prime p and all fixed k, r, the class of central extensions of $\mathbb{Z}_p^n$ by $\mathbb{Z}_p^k$ whose commutation form admits a layout of cut-rank at most r is uniformly automatic: a single finite automaton, reading a linear-size advice that spells out rank-r factorisations of the crossing blocks, decides the multiplication of every group in the class. The automaton state is exactly the cut-rank interface, so the parameter and the presentation coincide. Consequences include linear-time first-order model checking on all members, decidability of the uniform first-order theory of the class, and a common generalisation of several known automatic presentations: extraspecial p-groups, groups with commutation graphs of bounded pathwidth or treewidth, and—although its commutation graph is complete and has pathwidth $n-1$—the group in which no two generators commute. Generic forms have cut-rank $\Omega(n)$ under every layout, so the class is a thin, structured slice of the wild stratum. The constructions are implemented and machine-verified in the AutStr library. We leave open a converse in the spirit of Courcelle–Oum and the complexity of isometry for matrix spaces of bounded rank-width.

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Supplementary Material *Software (Source Code)*: `https://github.com/fariedabuzaid/AutStr`

1 Introduction

Two research programmes meet in this paper.

Algorithmic group theory. Among finite groups, the nilpotent groups of class 2 and exponent p are simultaneously the most numerous and the least classifiable. The Higman–Sims bounds show that the number of groups of order p^m is $p^{\frac{2}{27}m^3+O(m^{8/3})}$, and the lower bound is already attained by class-2 groups of exponent p [9, 17, 16]; it is a long-standing folklore conjecture that almost all finite groups are of this kind. By the Baer correspondence [1], such a group is determined by an alternating bilinear map $B\colon \mathbb{F}_p^n \times \mathbb{F}_p^n \to \mathbb{F}_p^k$, and its isomorphism problem is exactly the isometry problem for alternating matrix spaces, which is complete for the tensor-isomorphism class [6]. Model theory tells the same story from another side: by Mekler's construction, this variety of groups interprets all graphs and is as wild as first-order model theory permits [13]. In short, class-2 exponent-p groups are the dark matter of finite group theory: they form the bulk of the universe and resist classification.

Algorithmic model theory. Automatic structures [12, 2, 5] present a structure by finite automata: elements are words, and each relation is recognised by a synchronous multi-tape automaton. Automaticity buys decidable first-order theories and linear-time query evaluation. For a *class* of finite structures the right notion is *uniform* automaticity with advice [20, 18]: one tuple of automata presents every member of the class, and an additional advice string, read synchronously with the element encodings, selects the member. Uniformly automatic classes admit algorithmic meta-theorems in the style of Courcelle's theorem [3]: every first-order property can be compiled, once and for all, into an automaton that runs in linear time on each member's advice. Known uniformly automatic classes include all finite abelian groups, finite boolean algebras, and graphs of bounded tree-depth [18]. On the group-theoretic side the known automatic presentations stay close to commutativity: finitely generated word-automatic groups are virtually abelian [15], and the recent positive results of Nies and Stephan for infinite class-2 groups concern the infinite extraspecial group E_p, while the relatively free class-2 exponent-p group on infinitely many generators is *not* word-automatic [14].

This paper. We show that a single combinatorial parameter—the *rank-width of the commutation form*—carves a uniformly automatic slice out of the wild stratum, and that the parameter is not merely sufficient but is *literally* the automaton: the information a streaming algorithm must carry across a cut of the generator set is the crossing block of the form at that cut, and if that block factors through rank r, then r residues modulo p suffice.

Fix a prime p and let a *form* be a family $B = (B_{ji})_{1\le i<j\le n}$ of labels $B_{ji} \in \mathbb{F}_p^k$. It presents the group $G(B)$ generated by $x_1, \dots, x_n$ and central $y_1, \dots, y_k$ subject to $x_i^p = y_l^p = 1$ and the commutation relations $x_j x_i = x_i x_j\, y^{B_{ji}}$, a class-2 central extension of $\mathbb{Z}_p^n$ by $\mathbb{Z}_p^k$ of order p^{n+k}. For a set $S \subseteq [n]$ of generators, the *crossing block* M_S collects the labels of all commutation pairs with exactly one endpoint in S, and the *cut-rank* of S is $\mathrm{rk}_{\mathbb{F}_p} M_S$. A *linear layout* is an ordering of the generators; its width is the maximal cut-rank of an initial segment. A *tree layout* places the generators on a binary tree; its width is the maximal cut-rank of a subtree. For fixed k and r we prove:

- the class of all $(B, \text{linear layout of width} \le r)$ is uniformly word-automatic, with a multiplication automaton of essentially p^{k+r} states (Theorem 4);
- the class of all $(B, \text{tree layout of width} \le r)$ is uniformly tree-automatic, with essentially p^{k+2r} states (Theorem 10).

The advice has one letter per generator and spells out, cut by cut, a rank-r factorisation of the crossing block: a basis-change matrix, the coefficient of the newly read digit, and a read-off matrix that recovers the commutator corrections. Every well-formed advice string presents a member of the class, because the streamed cocycle is bilinear by construction; so the presentation is exact, not merely an over-approximation.

Three families illustrate the reach of the parameter and its independence from the usual width measures. A perfect matching of commutation pairs, all hitting one central generator, is the extraspecial group p^{1+2m}: width 1, recovering the finite counterpart of the Nies–Stephan presentation of E_p [14]. A commutation graph of bounded pathwidth or treewidth has bounded width (Proposition 12). But also the *complete* commutation graph—no two generators commute, pathwidth $n-1$, every vertex active at every cut—has width 1: its crossing blocks are all-ones matrices. An automaton that stores digits of active vertices cannot multiply in this group with constant memory; an automaton that stores the single functional $\sum_i a_i$ can, and does. This is the sense in which cut-rank, not vertex-based width, is the right invariant, mirroring the step from treewidth to rank-width and clique-width in graph theory [11, 4].

For the group theorist, the dividends are algorithmic meta-theorems on a stratum where classification fails: every first-order property of groups (commutativity, centrality of an element, existence of elements of a given order, and so on) compiles into an automaton that evaluates in time linear in $\log |G|$ on every member, uniformly across the class (Corollary 15); the uniform theory of the whole class is decidable (Corollary 16). Conversely, we are explicit about what the parameter costs in group-theoretic generality: for $k = 1$ and odd p the symplectic normal form collapses the class, up to isomorphism, to central products of Heisenberg groups with abelian factors, and bounded *vertex cover* of the commutation graph collapses to a bounded core times an elementary abelian factor (Proposition 13); genuinely new isomorphism types therefore live at $k \geq 2$, where the classification problem is wild and normal forms are unavailable—precisely where a presentation that works *without* normalisation earns its keep. A generic form has cut-rank $\Omega(n)$ under *every* layout (Proposition 14), so the class is thin, as any decidable slice of a wild stratum must be.

Both constructions are implemented in the AutStr library [19] and machine-verified: the multiplication automata are checked exhaustively against the group law on several layouts and forms, the word and tree presentations are checked against each other on shared layouts, and the first-order examples of this paper run as regression tests (Section 7).

Open problems. The two questions we consider most interesting are stated in Section 8: a *converse* in the spirit of Courcelle–Oum [4]—does uniform automaticity of a class of forms force bounded cut-rank under some layout?—and whether isometry of alternating matrix spaces of bounded rank-width is decidable in polynomial time, the matrix-space analogue of isomorphism testing for graphs of bounded rank-width [7].

2 Preliminaries

Words, trees, convolution. Σ denotes a finite alphabet and $\square \notin \Sigma$ a padding symbol. The *convolution* $w_1 \otimes \ldots \otimes w_m$ of words is the word over $(\Sigma \cup \{\square\})^m$ obtained by aligning the w_i position-wise and padding the shorter ones. A *tree* is a finite, binary, rooted, labelled tree; the convolution of trees overlays them and pads missing positions. A (bottom-up deterministic) tree automaton processes a tree from the leaves to the root; a run assigns a state to every node as a function of the node's label and the states of its children.

Uniformly automatic classes. We follow [20]. A *uniformly automatic presentation* of a class $\mathcal{C}$ of τ-structures consists of a regular advice set $A \subseteq \Sigma^*$ (or a regular set of advice trees) and automata $\mathfrak{U}, (\mathfrak{A}_R)_{R\in\tau}$ such that for every $\alpha \in A$ the structure

$$\mathfrak{S}_\alpha \;=\; \big(\{u : \mathfrak{U} \text{ accepts } \alpha \otimes u\},\; \big(\{\bar{u} : \mathfrak{A}_R \text{ accepts } \alpha \otimes \bar{u}\}\big)_{R\in\tau}\big)$$

is well-defined, and $\mathcal{C} = \{\mathfrak{S}_\alpha : \alpha \in A\}$ up to isomorphism. The same structure may be presented by several advices. The fundamental theorem of automatic structures relativises: for every first-order formula $\varphi(\bar{x})$ there is an automaton recognising $\{\alpha \otimes \bar{u} : \mathfrak{S}_\alpha \models \varphi(\bar{u})\}$, effectively [20]. We present groups in the signature $\{M, \mathrm{eq}\}$ with M the graph of multiplication.

Forms and groups. Fix a prime p and $k \geq 1$. A *form* of dimension n is a family $B = (B_{ji})_{1\leq i<j\leq n}$ with $B_{ji} \in \mathbb{F}_p^k$; we write $\hat{B}_{\{i,j\}} := B_{ji}$ for the unoriented label and treat absent

labels as 0. The group $G(B)$ is presented by generators $x_1, \ldots, x_n$, $y_1, \ldots, y_k$ and relations

$$x_i^p = y_l^p = [\,y_l, g\,] = 1 \quad \text{and} \quad x_j x_i = x_i x_j\, y^{B_{ji}} \ \ (i < j), \qquad \text{where } y^v := y_1^{v_1} \cdots y_k^{v_k}.$$

Define the bilinear map $C \colon \mathbb{F}_p^n \times \mathbb{F}_p^n \to \mathbb{F}_p^k$ by

$$C(a, a') \;=\; \sum_{1 \le i < j \le n} B_{ji}\, a_j\, a'_i. \tag{1}$$

▶ **Lemma 1** (normal forms). *$G(B)$ is isomorphic to the group with universe $\mathbb{F}_p^k \times \mathbb{F}_p^n$ and multiplication*

$$(b, a) \cdot (b', a') \;=\; \big(b + b' + C(a, a'),\ a + a'\big).$$

In particular $|G(B)| = p^{n+k}$ and every element has a unique normal form $x_1^{a_1} \cdots x_n^{a_n} y^b$.

Proof. Since C is bilinear, $C(a, a') + C(a + a', a'') = C(a, a') + C(a, a'') + C(a', a'') = C(a', a'') + C(a, a' + a'')$, which is the cocycle identity, so the displayed operation is associative; inverses are $(-b + C(a, a), -a) = (-b, -a)$ since $C(a, a)$ need not vanish—note however $C(e_i, e_i) = 0$ because B_{ii} is not among the labels, whence $(0, e_i)^p = (0, 0)$ also for $p = 2$. The elements $(0, e_i)$ and $(e_l, 0)$ satisfy all defining relations of $G(B)$, so $G(B)$ surjects onto the displayed group of order p^{n+k}; conversely the relations rewrite every word to a normal form, so $|G(B)| \le p^{n+k}$. ◀

▶ Remark 2 ($p = 2$). For odd p, $G(B)$ has exponent p. For $p = 2$ only the *generators* are involutions; central elements of order 4 may occur. The smallest example is the triangle $n = 3$, $k = 1$, all three labels 1: here $z = x_1 x_2 x_3$ is central with $z^2 = y$, and $G(B)$ is the 1-qubit Pauli group. This is a feature, not a bug: the presentation is consistent, and Lemma 1 holds verbatim.

By the Baer correspondence [1], isomorphism of the groups $G(B)$ (for odd p, modulo the radical) is isometry of the associated alternating matrix spaces; Hall's isoclinism [8] identifies exactly the data (n, B) modulo the radical. All parameters introduced next are therefore isoclinism invariants of the presented group, up to the choice of layout.

Cut-rank and layouts. For $S \subseteq [n]$ the *crossing block* of B at S is the matrix

$$M_S \in \mathbb{F}_p^{\left(([n]\setminus S)\times[k]\right)\times S}, \qquad (M_S)_{(j,l),\,i} \;=\; \big(\hat{B}_{\{i,j\}}\big)_l,$$

and the *cut-rank* of S is $\rho_B(S) := \operatorname{rk}_{\mathbb{F}_p} M_S$. A *linear layout* of B is an ordering of $[n]$; renaming generators, we always take it to be the natural order, and define its width as $\max_{1 \le t < n} \rho_B([t])$. A *tree layout* is a binary rooted tree T together with a bijection of $[n]$ onto the nodes of T; renaming, we take the bijection to be the post-order enumeration (left subtree, right subtree, node), so that every subtree hosts an interval $[\mathrm{lo}_v, v]$. Its width is $\max_v \rho_B(S_v)$ over the subtree generator sets S_v. We write $\mathrm{lcr}(B)$ and $\mathrm{tcr}(B)$ for the minimal width over all linear (tree) layouts. Since a chain is a tree, $\mathrm{tcr} \le \mathrm{lcr}$. For $k = 1$ and 0/1 labels, M_S is the bipartite adjacency matrix between S and its complement in the commutation graph, so tcr coincides with $\mathbb{F}_p$ rank-width (Proposition 11 makes the interface exact).

For fixed p, k, r let $\mathcal{C}_{p,k,r}^{\mathrm{lin}}$ denote the class of pairs (form, linear layout of width $\le r$), each presenting the group $G(B)$, and $\mathcal{C}_{p,k,r}^{\mathrm{tree}}$ the tree analogue.

3 Linear layouts: the word-automatic class

Throughout this section fix p, k, r and a form B with a linear layout of width $\le r$; write $M_t := M_{[t]}$.

The multiplication automaton must verify $(b_z, a_z) = (b_x, a_x) \cdot (b_y, a_y)$ while reading the three encodings synchronously, digit by digit. The digit sums $a_{z,t} = a_{x,t} + a_{y,t}$ are position-wise; the work is the correction $C(a_x, a_y) = \sum_t a_{x,t}\big(\sum_{i<t} B_{ti}\, a_{y,i}\big)$: at position t the automaton must know the vector $\sum_{i<t} B_{ti}\, a_{y,i} \in \mathbb{F}_p^k$, a linear functional of all digits read so far. It cannot store the digits; it stores their image under a factorisation of the crossing block.

▶ **Lemma 3** (streaming factorisation). *For $0 \le t \le n$ fix $V_t \in \mathbb{F}_p^{r\times t}$ whose rows span* $\operatorname{rowsp}(M_t)$ *(padding with zero rows; V_0 is empty). Then for every $t \in [n]$ there are $T_t \in \mathbb{F}_p^{r\times r}$, $v_t \in \mathbb{F}_p^r$, $R_t \in \mathbb{F}_p^{k\times r}$ with*

1. *$V_t = \big[\, T_t V_{t-1} \,\big|\, v_t \,\big]$, i.e. the first $t-1$ columns of V_t equal $T_t V_{t-1}$ and the last column is v_t;*
2. *$R_t V_{t-1} = B^{(t)}$, where $B^{(t)} \in \mathbb{F}_p^{k\times(t-1)}$ has columns B_{ti} for $i < t$.*

Proof. (1) Each row of M_t is indexed by some (j, l) with $j > t$; its restriction to the first $t-1$ columns is the corresponding row of M_{t-1} (the same generator j also lies outside $[t-1]$). Restriction is linear, so every row of V_t, being a combination of rows of M_t, restricts into $\operatorname{rowsp}(M_{t-1}) = \operatorname{rowsp}(V_{t-1})$; T_t collects the coefficients, and v_t is the last column. (2) The l-th row of $B^{(t)}$ is the row (t, l) of M_{t-1}, because $t \notin [t-1]$; hence it lies in $\operatorname{rowsp}(V_{t-1})$. ◀

▶ **Theorem 4** (linear layouts). *For all fixed p, k, r, the class $\mathcal{C}^{\mathrm{lin}}_{p,k,r}$ is uniformly word-automatic. The advice for (B, layout) has length $n + k$ over an alphabet of $p^{r^2+r+kr} + p + 2$ letters; the multiplication automaton has $p^{k+r} + \sum_{j<k} p^j + 1$ states; and every well-formed advice presents a member of the class.*

Proof. *Encoding.* An element (b, a) is the word $b_1 \cdots b_k a_1 \cdots a_n$ over $\{0, \dots, p-1\}$. The advice is $\mathsf{c}^k \alpha_1 \cdots \alpha_n$ where α_t encodes the triple (T_t, v_t, R_t) of Lemma 3 as $r^2 + r + kr$ digits.

The automaton. States are (c, j, d) with $j < k$, $d \in \mathbb{F}_p^j \times \{0\}^{k-j}$, and (x, d, w) with $d \in \mathbb{F}_p^k$, $w \in \mathbb{F}_p^r$, plus a sink. On the j-th center position it records $d_j := b_{z,j} - b_{x,j} - b_{y,j}$. On position t it checks $a_{x,t} + a_{y,t} = a_{z,t}$, updates

$$d \;\leftarrow\; d - a_{x,t} \cdot (R_t\, w), \qquad w \;\leftarrow\; T_t\, w + v_t \cdot a_{y,t},$$

and accepts at the end iff $d = 0$.

Correctness. By induction, after position t the invariant $w = V_t\,(a_y)|_{[t]}$ holds: it is preserved by Lemma 3(1). Hence at position t, $R_t w = R_t V_{t-1}(a_y)|_{[t-1]} = \sum_{i<t} B_{ti} a_{y,i}$ by Lemma 3(2), so the accumulated subtraction is exactly $C(a_x, a_y)$ of (1), and acceptance means $b_z = b_x + b_y + C(a_x, a_y)$ as required by Lemma 1. The universe and equality automata only check the shape.

Exactness. Conversely, an arbitrary well-formed advice defines, through the same update equations, the correction $\tilde{C}(a, a') = \sum_t a_t\, R_t\big(\prod\text{-terms in } T, v\big)a'$, which is bilinear in (a, a') and supported on pairs $i < t$; so it equals (1) for the form $B_{ti} := R_t T_{t-1} \cdots$ (coefficient extraction), and the advice presents $G(B)$ for that form, whose layout has width $\le r$ because the functional carried across every cut factors through $\mathbb{F}_p^r$ by construction. ◀

▶ Remark 5. The presentation is honest about sizes: the encoding of an element of G has length $n + k = \log_p |G|$, and for fixed φ the model-checking run (Corollary 15) is linear in $\log |G|$—not in $|G|$.

▶ **Example 6** (the clique). Let $B_{ji} = 1 \in \mathbb{F}_p^1$ for all $i < j$: no two generators commute, and the commutation graph is complete, with pathwidth $n-1$. Every crossing block is an all-ones matrix, of rank 1; the layout data can be taken as $T_t = (1)$, $v_t = (1)$, $R_t = (1)$, and the automaton degenerates to two registers: $w = \sum_{i \le t} a_{y,i}$ and the deficit d. Under any vertex-based width measure this is the hardest instance; under cut-rank it is the easiest.

▶ **Example 7** (extraspecial groups). Let $n = 2m$, $k = 1$, and $B_{2t,2t-1} = 1$ for $t \in [m]$: a perfect matching, presenting the extraspecial-type group of order p^{1+2m} (for odd p, the exponent-p extraspecial group; for $p = 2$, a central product of dihedral and quaternion groups). Any cut splits at most one pair, so the natural layout has width 1. This is the finite-class counterpart of the word-automatic presentation of the infinite extraspecial group E_p [14], which is the direct limit of this family.

4 Tree layouts: the tree-automatic class

Tree layouts strictly extend linear ones (a chain is a tree), and for $k = 1$ they capture rank-width (Proposition 11). The new difficulty is that a bottom-up automaton meets a crossing pair $\{i, j\}$ in one of *two* ways. Let S be the current subtree, with $i \in S$. If j is a proper ancestor of S's root, then—as in the linear case—the contribution $B_{ji}\, a_{x,j}\, a_{y,i}$ pairs a *future* x-digit against *past* y-digits. But if j lies in a sibling subtree, the two subtrees are processed independently and their contributions must be reconciled at the merge. In post-order the left subtree L precedes the right subtree R entirely, so every split pair has its smaller endpoint in L and larger in R, and contributes $B_{ji}\, a_{x,j}\, a_{y,i}$ with $i \in L$, $j \in R$: the *right* child's x-digits pair against the *left* child's y-digits. The automaton therefore carries *two* functional vectors, $w^x = V_S\,(a_x)|_S$ and $w^y = V_S\,(a_y)|_S$, and the merge is a bilinear pairing between them, which itself must factor through the two children's bases. That it does is a two-sided factorisation:

▶ **Lemma 8** (two-sided factorisation). *Let $X \in \mathbb{F}_p^{m \times m'}$, $V \in \mathbb{F}_p^{r \times m'}$, $W \in \mathbb{F}_p^{r \times m}$. If every row of X lies in* $\mathrm{rowsp}(V)$ *and every column of X lies in* $\mathrm{colsp}(W^\top)$*, then $X = W^\top Q\, V$ for some $Q \in \mathbb{F}_p^{r \times r}$.*

Proof. Write $X = AV$ where, w.l.o.g., the nonzero rows of V are linearly independent and A uses only those; then V has a right inverse Y on its row space (VY is the identity on the used coordinates), so $A = XY$, and every column of A is a combination of columns of X, hence lies in $\mathrm{colsp}(W^\top)$; write $A = W^\top Q$. ◀

▶ **Lemma 9** (tree streaming factorisation). *Fix a tree layout of B of width $\le r$, and for every node v with subtree interval S_v fix $V_v \in \mathbb{F}_p^{r \times |S_v|}$ whose rows span* $\mathrm{rowsp}(M_{S_v})$*. Then every node admits letter data over $\mathbb{F}_p$:*

- leaf v*: the column $v_v := V_v \in \mathbb{F}_p^{r \times 1}$;*
- unary v *with child c: T with $TV_c = V_v|_{S_c}$, the last column v_v, and R with $R\,V_c = B^{(v)}|_{S_c}$;*
- binary v *with children L, R: T_L, T_R and v_v as above per child block, read-offs R_L, R_R with $R_\bullet V_\bullet = B^{(v)}|_{S_\bullet}$, and $Q_1, \ldots, Q_k \in \mathbb{F}_p^{r \times r}$ with $V_R^\top Q_l\, V_L = X_l$, where $(X_l)_{j,i} = (B_{ji})_l$ for $i \in S_L$, $j \in S_R$.*

Proof. The T- and R-systems are solvable by the restriction argument of Lemma 3: rows of M_{S_v} restricted to a child block are rows of that child's crossing block, and the row (v, l) of the read-off lies in the child's crossing block since $v \notin S_c$. For Q_l: the row of X_l indexed by $j \in S_R$ is the row $((j, l), \cdot)$ of M_{S_L} restricted to nothing (it is exactly such a row), hence lies in $\operatorname{rowsp}(V_L)$; the column of X_l indexed by $i \in S_L$ is the row $((i,l),\cdot)$ of M_{S_R} transposed, hence lies in $\operatorname{colsp}(V_R^\top)$. Apply Lemma 8. ◀

▶ **Theorem 10** (tree layouts). *For all fixed p,k,r, the class $\mathcal{C}^{\text{tree}}_{p,k,r}$ is uniformly tree-automatic. The advice tree has one letter per generator plus a chain of k center markers above the root; the multiplication automaton has $p^{k+2r} + \sum_{j=1}^{k} p^{k-j} + 1$ states; every well-formed advice presents a member; and on chain-shaped layouts the presentation restricts to that of Theorem 4.*

Proof. Elements (b,a) are digit labellings of the advice's shape: digit a_v at the node of generator v, and the center digits $b_1,\dots,b_k$ on the marker chain, innermost first. The automaton's layout states are $(\mathsf{t}, s, w^x, w^y) \in \{\mathsf{t}\} \times \mathbb{F}_p^k \times \mathbb{F}_p^r \times \mathbb{F}_p^r$ with the invariant, at each node v:

$$s = \sum_{\substack{i<j \\ i,j\in S_v}} B_{ji}\, a_{x,j}\, a_{y,i}, \qquad w^x = V_v\,(a_x)|_{S_v}, \qquad w^y = V_v\,(a_y)|_{S_v}.$$

At a leaf, $s = 0$ and $w^\bullet = v_v \cdot a_{\bullet,v}$. At a unary node, the pairs internal to S_v but not to S_c are exactly $\{v,i\}$, $i \in S_c$; their contribution is $a_{x,v} \cdot (R\, w^y)$ by Lemma 9, and both functional vectors update as $w \mapsto Tw + v_v \cdot (\text{own digit})$. At a binary node the new pairs are the two read-offs plus the split pairs, whose contribution is

$$\sum_{i\in S_L,\, j\in S_R} B_{ji}\, a_{x,j}\, a_{y,i} \;=\; \Big((w_R^x)^\top Q_l\, w_L^y\Big)_{l\in[k]}$$

by Lemma 9; the functional vectors merge as $w \mapsto T_L w_L + T_R w_R + v_v \cdot (\text{own digit})$. Position-wise digit sums $a_{x,v} + a_{y,v} = a_{z,v}$ are checked at every node. At the root of the layout, $s = C(a_x, a_y)$; the k center markers then verify $b_{z,j} - b_{x,j} - b_{y,j} = s_j$ coordinate by coordinate, and the automaton accepts iff all coordinates match. Exactness is as in Theorem 4: the streamed correction is bilinear for every well-formed advice. On a chain (left spine) there are no binary nodes; w^x is dead weight and the run is the word automaton's. ◀

▶ **Proposition 11** (rank-width interface). *Extend tree layouts by* structural nodes *carrying no generator (the element encodings carry a forced* 0 *digit there; the automata are unchanged). Then for $k = 1$: a form belongs to $\mathcal{C}^{\text{tree}}_{p,1,r}$ (with structural nodes) if and only if the $\mathbb{F}_p$-labelled commutation graph of B has rank-width at most r over $\mathbb{F}_p$, and a rank-decomposition converts to a layout without changing the width.*

Proof. A rank-decomposition is a subcubic tree with the vertices at its leaves; rooting it at an arbitrary edge yields a binary tree whose internal nodes become structural. The generator set of any subtree is one side of an edge cut of the decomposition, and its crossing block is the bipartite adjacency block whose rank defines cut-rank; conversely a layout is such a decomposition after suppressing structural nodes. ◀

5 The class seen from group theory

We collect what bounded cut-rank does and does not mean for the groups themselves. Proofs in this section are standard linear algebra over $\mathbb{F}_p$ and are included for completeness.

▶ **Proposition 12** (classical width measures). *Let Γ be the commutation graph of B (edges = non-commuting pairs).*

1. *If* Γ *has a vertex cover of size* c*, every layout compatible with any order has width* $\leq c$*; in particular* $\operatorname{lcr}(B) \leq c$.
2. *If* Γ *has pathwidth* w*, then* $\operatorname{lcr}(B) \leq w + 1$.
3. *If* Γ *has treewidth* w*, then* $\operatorname{tcr}(B) \leq k(w + 1)$.

Proof. (1) Every nonzero entry of a crossing block lies in a row or column indexed by a cover vertex; at most c of these meet each side, and a matrix whose nonzero entries lie in c_1 columns and $c_2 k$ rows has rank $\leq c_1 + c_2 \cdot \min(k, \cdot) \leq c$ when counted per cover vertex. (2) Order the vertices by a path decomposition; the earlier endpoints of crossing edges at any cut are active vertices, so nonzero columns number $\leq w + 1$. (3) Expand each bag node of a tree decomposition into a chain introducing its new vertices, and take this as the layout. The bag of the current edge of the decomposition separates the subtree from the rest, so every crossing pair has an endpoint in it: nonzero columns for below-endpoints ($\leq w + 1$) and k rows per above-endpoint bound the rank by $k(w + 1)$. ◀

▶ **Proposition 13** (collapse at the tame end)**.** *Let* p *be odd.*

1. *If* Γ *has a vertex cover of size* c*, then* $G(B) \cong H \times \mathbb{Z}_p^m$ *with* $|H| \leq p^{2c+k}$*; the class of such groups is, up to isomorphism, a finite list of cores times elementary abelian factors.*
2. *For* $k = 1$*, every* $G(B)$ *is isomorphic to (extraspecial of exponent* p*)* $\times$ *(elementary abelian): the symplectic normal form aligns every form with a matching.*

Proof. (1) All nonzero entries of $\hat{B}$ lie in c rows and columns, so the total rank of the alternating map is $\leq 2c$; split off the radical (for odd p its preimage is elementary abelian and complements off the commutator support). (2) is the classical normal form of an alternating form: rank $2m$, matching of m hyperbolic pairs, radical split off. ◀

The reading for the mixed audience: at $k = 1$ (and bounded vertex cover for any k) the parameter buys presentation-theoretic rather than isomorphism-theoretic generality—the clique of Example 6 is abstractly a Heisenberg group times an abelian group, but the automaton multiplies in the *given* presentation without computing the normal form. The isomorphism-theoretic content of the class lives at $k \geq 2$, where pairs of alternating forms already have infinitely many normal-form families and triples are wild [6]; there, bounded cut-rank is a genuine structural restriction that does not reduce to a classical one. And the restriction is necessary in a precise sense:

▶ **Proposition 14** (generic forms are wide)**.** *Let* $k = 1$ *and let* B *be uniformly random. With high probability,* every *linear and every tree layout of* B *has width* $\geq n/2 - O(\sqrt{n})$*; that is,* $\operatorname{tcr}(B) = \Omega(n)$.

Proof sketch. For a fixed set S with $|S| = m \leq n/2$, the crossing block is a uniform random $(n - m) \times m$ matrix over $\mathbb{F}_p$, and $\Pr[\mathrm{rk} < m - s] \leq p^{-s(n-m-m+s)} \leq p^{-s^2}$. A union bound over all $\leq 2^n$ sets with $s = c\sqrt{n}$ leaves all sets of size $m \in [n/3, n/2]$ with rank $\geq m - c\sqrt{n}$. Every linear layout has a cut of size $\lfloor n/2 \rfloor$, and every binary tree has a subtree whose interval size lies in $[n/3, 2n/3]$; complement if necessary. ◀

Together with the Nies–Stephan negative result—the relatively free class-2 exponent-p group on ω generators, whose finite quotients have forms of full generic complexity, is not word-automatic [14]—this locates the constructions of Section 3 and Section 4 plausibly close to the true boundary of automaticity; Section 8 makes the conjecture precise.

6 Algorithmic consequences

The generic machinery of uniformly automatic classes [20] yields, for both presentations:

▶ **Corollary 15** (compiled model checking)**.** *For every first-order formula $\varphi(x_1, \dots, x_m)$ in the signature of groups there is (effectively) an automaton $\mathfrak{A}_\varphi$ such that for every advice $\alpha \in A$ and all elements $u_1, \dots, u_m$ of G_α:*

$$G_\alpha \models \varphi(\bar{u}) \iff \mathfrak{A}_\varphi \text{ accepts } \alpha \otimes u_1 \otimes \dots \otimes u_m.$$

In particular, once φ is compiled, deciding it on a member of the class takes time $O(n+k) = O(\log |G|)$.

▶ **Corollary 16** (decidable uniform theory)**.** *The advice sets of Theorem 4 and Theorem 10 are regular. Hence the uniform first-order theory $\{\varphi : G_\alpha \models \varphi$ for all $\alpha \in A\}$ is decidable, and so are existential class queries ("is there a member satisfying φ?"). The same holds for extensions of first-order logic by regularity-preserving quantifiers, e.g. cardinality and Ramsey quantifiers [20].*

▶ **Example 17.** The center is definable, $\mathrm{Cen}(x) \equiv \forall u \forall v \forall w\, (M(x,u,v) \wedge M(u,x,w) \rightarrow v = w)$, so Corollary 15 decides centrality of a given element in time $O(\log |G|)$ on every member; likewise commutativity of the member, existence of central elements of order 4 at $p = 2$ (distinguishing the Pauli triangle from the dihedral matching), or any fixed isoclinism-flavoured first-order invariant. One automaton per property serves the entire class.

7 Implementation and machine verification

Both constructions are implemented in the open-source library AutStr [19] as the classes `CutRankGroups` (p, k, r; Theorem 4) and `CutRankTreeGroups` (Theorem 10), together with layout compilers that realise Lemma 3 and Lemma 9 by Gaussian elimination over $\mathbb{F}_p$ and fail exactly when a cut exceeds the width bound. The test suite verifies, among other things: exhaustive agreement of the multiplication automata with the group law of Lemma 1 on spine and balanced layouts for the zero, complete, matching and star forms; agreement of the word and tree presentations on shared (chain) layouts; the complete form at $n = 14$ through the constant-size automaton; the Pauli triangle including first-order definable centrality; and the width guards (a form of cut-rank 2 is rejected at $r = 1$ and handled at $r = 2$). The first-order examples of Section 6 run through the library's generic query evaluation. We regard the implementation as part of the evidence: the solvability claims of Lemma 3 and Lemma 9 are exercised on every compiled layout.

8 Open problems

1. *Converse / characterisation.* Does uniform (word-, tree-) automaticity of a class of forms with fixed k force bounded (linear) cut-rank under suitably chosen layouts? A positive answer would be a Courcelle–Oum-style characterisation [4] of automaticity on the class-2 stratum; Proposition 14 and the negative result of [14] are the supporting evidence.
2. *Isometry in P.* Is isometry of alternating matrix spaces of bounded $\mathbb{F}_p$ rank-width decidable in polynomial time? This is the matrix-space analogue of isomorphism testing for graphs of bounded rank-width [7] and would identify a tame island inside a tensor-isomorphism-complete problem [6]. Note that computing a witnessing layout is itself nontrivial; for $k = 1$ rank-decompositions are FPT-computable [10].

3. *Unbounded centers.* The library contains a uniformly tree-automatic class of class-2 groups with *growing* k and laminar commutator targets (tree-indexed extraspecial groups), which is *not* subsumed by Theorem 10; its cuts also carry one-dimensional interfaces. A common generalisation—cut-rank of the full cocycle data with distributed central targets—seems within reach.
4. *Wider widths in practice.* The advice letters of Theorem 10 grow as $p^{\Theta(kr^2)}$; the implementation therefore bounds flat alphabets. Factored transition labels (shared BDDs) would make tree widths $r \geq 2$ practical.

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